# INTERIM FEDERAL SPECIFICATION

## CLEANING COMPOUND, ALKALI-TYPE

**This interim Federal specification has been developed by the Department of the Army-Ordnance Corps, based upon currently available technical information. It has not been approved for promulgation as a Federal specification and is subject to modification. This specification is in interim form only, and, pending its promulgation by the Administrator, General Services, as a regularly coordinated Federal specification, it may be used in procurement.**

1. SCOPE

1.1 Scope.- This specification covers one class of alkaline cleaning compounds suitable for use in the hot soak tank cleaning of ferrous and non-ferrous alloy parts.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

Federal

| | | |
|---|---|---|
| O-S-604 | - | Sodium Metasilicate, Pentahydrate. |
| O-T-671 | - | Trisodium Phosphate, Technical (Phosphate Cleaner). |
| R-T-143 | - | Tars (For use in) Road Construction. |
| QQ-A-561 | - | Aluminum; Plates and Sheets(AL-2). |
| SS-A-706 | - | Asphalt; (For use in) Road and Pavement Construction. |

Military

MIL-L-15016 - Lubricating Oil; General Purpose.

STANDARD

MIL-STD-129 - Marking of Shipments.

(Copies of Federal specifications may be obtained upon application, accompanied by postal note, money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

2.2 Specifications and other publications applicable only to individual departments are listed in section 7.

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3. REQUIREMENTS

3.1 Qualification (type or brand approval).- Unless otherwise specified, cleaning compound, alkali-type purchased under this specification shall be of brand or type which has been tested and found to meet all the requirements of section 3 of this specification. (See 6.4).

3.2 Materials.- The raw materials used in manufacture of these compounds shall be of high quality, intimately assembled and processed so as to produce a granular product which will remain uniform in composition and show no evidence of segregation or caking during handling or storage. The compound when used as a hot soak cleaner at boiling temperature and a concentration of 7.6 percent weight to volume shall effectively remove tar, asphalt, mineral oil, grease and road dirt from metal parts. Non-ferrous metals shall not be attacked under these conditions.

3.3 Composition.- The cleaning compound is not required to conform to definite chemical composition requirements. The manufacturer is given wide latitude in the selection of raw material and processes of manufacture, provided that the product meets all applicable requirements of this specification. The compound shall not contain any toxic or volatile substances, such as chlorinated hydrocarbons, phenolic compounds, benzene, or the like.

3.4 Performance requirements.- The cleaning compound, alkali-type, shall be equal or superior in effectiveness to the comparison formula in 4.4.1.3. in all respects when tested as specified in 4.4.1. The compound covered by this specification shall be tested simultaneously with the comparison alkali-type cleaning compound specified in 4.4.1.3.

3.4.1 Cleaning efficiency.- The cleaning compound covered by this specification shall be equal or superior to the standard comparison cleaning compound specified in 4.4.1.3, prepared and tested under the same conditions, in ability to remove mineral oil, tar, and asphalt soils from metal surfaces when tested as specified in 4.4.1.1.

3.4.2 Stability.- The compound covered by this specification, when boiled for 40 hours and tested as specified in 4.4.1.2, shall be equal or superior to the standard comparison formula, prepared and tested under the same conditions, in ability to remove soils from metal surfaces.

3.5 Corrosiveness.- The cleaning compound, alkali-type, shall not attack aluminum. When tested in accordance with 4.4.2, there shall be no loss in weight of aluminum test panels conforming to Specification QQ-A-561.

3.6 pH Value.- The pH value of a distilled water solution of the compound at a concentration of 7.6 percent weight to volume shall be not more than 12.1 when tested at 25°C, as specified in 4.4.3.

3.7 Dust forming properties.- The dust forming properties shall be such that, when tested as specified in 4.4.4, the dust shall settle within a period of five seconds.



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3.8 Workmanship.- The compound shall be manufactured in accordance with the best commercial practice to produce a high quality material which is stable and not subject to change during storage in a sealed container.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Sampling.

4.1.1 Lot.- For purposes of sampling a lot shall consist of all material mixed or prepared at one time in one vessel. Material shall be identified by order of production or batch number.

4.1.2 Sampling procedure.- A minimum of two samples, each weighing approximately one (1) pound, shall be taken from each lot.

4.2 Inspection.- Inspection shall be at the point of delivery, unless otherwise specified.

4.3 Tests.- Testing under this specification shall be for the purpose of:

a. Qualification.
b. Acceptance of individual lots.

4.3.1 Qualification tests.- Qualification tests shall consist of tests for all the requirements of this specification. Qualification tests shall be conducted at a Government Laboratory designated by the Ordnance Corps, Department of the Army(see 6.4).

4.3.2 Acceptance tests.- Inspection tests for acceptance of individual lots shall be made at a Government laboratory and shall normally consist of tests for all requirements specified in section 3 with the exception of the cleaning efficiency and stability tests for mineral oil and asphalt. (The cleaning efficiency and stability tests for inspection purposes shall normally consist of tests on tar soils only.) The procuring agency, however, reserves the right to make the tests for compliance to all the requirements specified above.

4.4 Test methods.-

4.4.1 Performance tests.- Performance tests shall consist of:

a. Cleaning efficiency.
b. Stability.

4.4.1.1 Cleaning efficiency test.-

4.4.1.1.1 Preparation of test panels.- Test panels of WD-1020, 18-20 gauge, cold-rolled steel, 2-½ by 2-½-inches in size, with a ¼-inch diameter hole placed 1/8-inch from one corner, shall be prepared. Sharp edges shall be smoothed with No. 1 emery cloth. The panels shall then be degreased in C.P. grade acetone. Polish both faces with No. 1 emery cloth stroking in one direction only. Clean in a hot alkaline solution until free from water break. Rinse in water, then dip in absolute ethyl alcohol, and wipe dry with paper toweling.

4.4.1.1.2 Application of soils.- The following soils shall be applied to separate panels:

4.4.1.1.2.1 Mineral oil.- The soiling material shall be Military Symbol Oil 3065 conforming to Specification MIL-L-15016A.



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The cleaned panels are suspended on an "S" hook and dipped into a 400-ml beaker of the oil at a temperature of 26°± 1°C. They are then removed and allowed to drain at a temperature of 26°± 1°C. for 30 minutes. The remaining drop of oil in the lower corner of each panel is then removed with absorbent cotton. The amount of mineral oil adhering per panel shall be approximately 0.16 gram.

4.4.1.1.2.2 Tar.- The soiling material shall be tar conforming to Specification R-T-143, grade RT-11. The soiling material is spread evenly over one face of the cleaned test panels using the edge of a steel spatula, leaving unsoiled the triangular area at the corner which includes the ¼-inch hole. The amount of tar on each panel shall be 0.32-0.34 g., determined by trial and error. Allow each panel to stand for 15 minutes before stripping.

4.4.1.1.2.3 Asphalt.- The soiling material shall be petroleum asphalt conforming to Specification SS-A-706, grade 85-100, designation AP-3. The soil, heated to approximately 50°C., is spread evenly over one face of the cleaned test panels using the edge of a steel spatula, leaving unsoiled the triangular area at the corner which includes the ¼-inch hole. The amount of asphalt on each panel shall be 0.32-0.34 g., determined by trial and error. Allow each panel to stand for 15 minutes before stripping.

4.4.1.1.3 Preparation of cleaning solutions.- In a 2-liter beaker, prepare 1600 ml. of a 7.6 percent, weight to volume, distilled water solution of the test compound. A similarly prepared solution shall be made of the standard comparison compound. Bring solutions to a boil and keep at this temperature throughout the test. Maintain solution level throughout the test by addition of distilled water.

4.4.1.1.4 Cleaning procedure.- Three test panels for each type of soil, prepared as specified in 4.4.1.1.2, shall be cleaned successively as specified hereinafter in the following order: tar, mineral oil, then asphalt. The soiled test panel is immersed in the cleaning solution specified in 4.4.1.1.3 by means of an iron hook, one end of which passes through the 1/4-inch hole and the other end over a glass rod placed across the top of the 2-liter beaker. At the end of 3-minute intervals the test panel is moved forward and backward, six times in each direction, and then up and down three times in each direction. This agitation of the panels is applied in the cleaning of the tar and asphalt test panels but not in the cleaning of the mineral oil panels. The panel is then removed from the solution for visual inspection of the extent of soil removal and quickly replaced. The entire operation of agitation of the test panel and visual inspection shall require not more than 6 seconds. Mineral oil test panels shall be cleaned for 3 minutes, tar-soiled panels for 39 minutes and asphalt-soiled panels for 21 minutes respectively. At the end of the specified cleaning period, or before



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if visual inspection indicates completeness of soil removal, the panel shall be removed from the cleaning solution and given 2 six-second rinses in distilled water (no agitation); there shall be a four-second drain between rinses. The rinsing solutions shall consist of a 2 one-liter beakers, each containing 800 ml. of distilled water at 23°±3°C.

4.4.1.1.5 Measurement of residual soil.

4.4.1.1.5.1 Tar and asphalt.- Rinsed tar and asphalt test panels, after treatment specified in 4.4.1.1.4., are dried for 15 minutes in an oven at a temperature of 75°±1°C., followed by cooling at room temperature for 30 minutes. They are then examined for number and size of residual spots of soil, and weighed to the nearest 0.0001 g. The residual tar or asphalt is scrubbed off with benzene or xylene, using paper toweling, after which the panel is rinsed in absolute ethyl alcohol, dried with paper toweling, and reweighed. The difference in weight is the amount of residual soil. Tests shall be run in triplicate.

4.4.1.1.5.2 Mineral oil.- Rinsed mineral oil test panels, after treatment as specified in 4.4.1.1.4., shall be examined for presence or absence of water-break. They are then dried for 15 minutes at a temperature of 75°±1°C., followed by cooling at room temperature for 30 minutes. They are then examined for presence or absence of residual stains ("residue pattern" stains), the number and approximate size of which are recorded. Tests shall be run in triplicate.

4.4.1.1.6 Cleaning efficiency.- The compound covered by this specification shall be equal or superior in cleaning efficiency to the standard comparison cleaner specified in 4.4.1.3., prepared and tested under the same conditions, in soil removal as measured by the weight of residual soil. Tests shall be run in triplicate for each type of soil. (See 6.3).

4.4.1.2 Stability test.-

4.4.1.2.1 Preparation of solutions.- In a 3-liter beaker, prepare 1600 ml. of a 7.6 percent, weight to volume, distilled water solution of the test compound. Boil the solution for 40 hours, replacing water lost by evaporation as necessary. Transfer the solution to a 2-liter beaker, bring to a boil, and keep the solution at a boil throughout the test, maintaining the solution level by addition of distilled water.

4.4.1.2.2 Stability test procedure.- Using the solutions prepared in 4.4.1.2.1, three test panels for each type of soil prepared and soiled in accordance with 4.4.1.1.1, and 4.4.1.1.2, shall be cleaned successively as specified in 4.4.1.1.4 (a total of nine test panels). The residual solution on each panel shall be determined in accordance with 4.4.1.1.5. The test panels shall be cleaned in the following order: tar, mineral oil, then asphalt.

4.4.1.3 Standard comparison cleaning compound.- The composition of the standard comparison cleaning compound shall be as follows:



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| Component | Percent by Weight |
|---|---|
| Sodium metasilicate, pentahydrate | 34.5 |
| Sodium phosphate, monobasic | 12.0 |
| Trisodium phosphate, dodecahydrate | 33.5 |
| Non-ionic surface-active agent | 5.2 |
| Anionic surface-active agent | 14.8 |

The components of the standard comparison alkali-type cleaning compound shall conform to the following requirements:

1. Sodium metasilicate, pentahydrate---Specification O-S-604.
2. Trisodium phosphate dodecahydrate---Specification O-T-671.
3. Sodium phosphate monobasic---C.P. grade $NaH_2PO_4.H_2O$.
4. Non-ionic surface-active agent..."Triton X-100" manufactured by Rohm and Haas, "Igepal CA-Extra High Concentration" manufactured by General Dyestuff Corporation, or equal.
5. Anionic surface-active agent..."Nacconal NR" manufactured by National Aniline Division, Allied Chemical & Dye Corporation, Santomerse #3 manufactured by Monsanto Chemical Company, or equal.

4.4.2 Corrosion test.

4.4.2.1 Preparation of test panels.- Test panels shall be cut 3 inches by 3/4 inch in size from 1/8 inch thick sheet aluminum conforming to Specification QQ-A-561, ½ H temper. Sharp edges shall be smoothed with No. 1 emery cloth. Polish both faces with the same abrasive, stroking in one direction only.

4.4.2.2 Cleaning of panels.- Panels shall be cleaned with C. P. acetone using a swab of absorbent cotton. They shall then be wiped with paper toweling, dipped in absolute ethyl alcohol, and again wiped with paper toweling.

4.4.2.3 Corrosion test procedure.- The clean test panel shall be weighed to nearest 0.0001 gram and immersed completely in 200 ml. of a 7.6 percent, weight to volume, distilled water solution of the compound, then boiled utilizing a 500-ml. tall-form beaker. The solution shall be kept at a boil throughout the test, the solution level being maintained by additions of distilled water as required. After 60 minutes, the test panel shall be removed from the boiling solution, rinsed under flowing tap water (cold), rinsed in distilled water (20° - 26°C.), dipped in absolute ethyl alcohol, wiped dry with paper toweling, and reweighed. The presence of corrosion products shall be noted. Tests shall be run in duplicate.

4.4.3 pH value.-

4.4.3.1 Apparatus.- The test shall be made with a pH electrometer having a sensitivity and readability of at least 0.05 pH. A sealed type, alkali-resistant glass electrode shall be used with a calomel reference electrode. The meter shall be standardized against a pH 10 buffer solution immediately before making a test. The instrument to be used shall be the Beckman pH meter Laboratory Model H or equal and the electrode shall be the Beckman "Type E" high glass electrode or equal.



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4.4.3.2 Procedure.- Prepare 100 ml. of a 7.6 percent, weight to volume, distilled water solution of the compound. Determine pH at 25°C. No correction for sodium ion concentration shall be made.

4.4.4 Dust forming.- The inside walls of a 250-ml. glass stoppered graduate shall be rendered completely free of any grease and moisture. A 25-gram sample of the cleaner shall then be placed on the bottom of the clean tube. The tube shall be then stoppered, inverted and immediately returned to its original position. After any suspended dust has been allowed to settle for 5 seconds, the stopper of the tube shall be removed and a moistened piece of red litmus paper suspended in the tube without touching the walls so that the lower end of the litmus paper strip coincides with the 210 ml. mark at the upper end of the graduate. After 60 seconds immersion, the litmus paper shall be removed and examined for any color change.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Unless otherwise specified, commercial packages are acceptable under this specification.

5.2 Packing.- Unless otherwise specified, the subject material shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carrier, at the lowest rate, to the point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Individual containers.- Unless otherwise specified, each container shall be durably and legibly marked as follows:

"Cleaning Compound, Alkali-Type
Specification P-C-436(Army-Ord)
Stock Number_____
Quantity contained_____Date of manufacture_____
Name of contractor_____Batch or Lot No._____
Government Order or Contract No._____."

5.3.2 Shipment.- In addition, all shipments shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- Cleaning compound, alkali-type, covered by this specification is intended for use in the hot soak cleaning of ferrous and non-ferrous alloy parts.

This specification is intended to supersede Ordnance Corps Tentative Specification AXS-1849, Compound, Cleaning, Alkali-Type.

6.2 Ordering date.- Procurement documents should specify the title, number, and date of this specification.

6.3 Degree of cleanliness obtained with Standard Comparison Cleaner.- It has been found that test panels, soiled and cleaned with the standard comparison formula as specified herein, have



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shown the following results:

Tar soil....After 39 minutes, not more than 1.0 mg. of residual soil in the form of scattered spots no larger than 1/32-inch in any dimension.

Asphalt soil....After 21 minutes, the test panel is completely cleaned.

Mineral oil soil....After 3 minutes, the test panel shows no water-break and no "residue-pattern" stains.

6.4 In the procurement of products requiring qualification the right is reserved to reject bids on products that have not been subjected to the required tests and found satisfactory for inclusion on the Federal Qualified Products List. The attention of suppliers is called to this requirement, and manufacturers are urged to have the products that they propose to offer to the Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products covered by this specification may be obtained by communicating with: The Commanding General, Aberdeen Proving Ground, Maryland, Attention: Paint and Chemical Laboratory.

Notice.- When Government drawings, specifications or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following departmental specifications, and standards of the issue in effect on date of invitation for bids, form a part of this specification for purchases made under this specification by the respective departments.

7.1.1 Army, Navy and Air Force.-

STANDARDS

MIL-STD-129 - Marking of Shipments.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)



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